

Structural Analysis of Evolutionary Collapse and Genetic Residue under Exogenous Shocks

Abstract

Evolutionary outcomes are often interpreted as deterministic consequences of adaptive superiority. However, such interpretations frequently rely on retrospective coherence rather than derivability from prior system dynamics. This paper presents a structural analysis of evolutionary collapse under exogenous shocks, using the Last Glacial Maximum (LGM) as a case study. We argue that the extinction of Neanderthals and the persistence of Neanderthal genetic material within modern Eurasian populations constitute evidence of a non-derivable bifurcation rather than a competitive replacement. The genetic residue observed in contemporary humans functions as a structural fossil: a trace of an evolutionary trajectory that collapsed as a system but survived as absorbed information. This analysis challenges deterministic narratives in evolutionary theory and reframes human genetic history as the outcome of contingent structural bifurcations imposed by external constraints.

“This analysis applies a structural framework of non-derivable bifurcation, developed in prior work on the Theory of Axiomatic Necessity (TNA), to evolutionary history.”

“This paper adopts a structural rather than population-genetic modeling approach.”

1. Introduction

In evolutionary discourse, outcomes are commonly explained through narratives of adaptation, competition, and fitness optimization. Extinction events are retrospectively framed as inevitable consequences of inferiority, while survival is interpreted as proof of superiority. Such explanations often conflate *intelligibility after the fact* with *derivability before the event*.

This paper challenges that conflation.

We propose that certain evolutionary outcomes are not the result of continuous adaptive trajectories, but rather of **structural collapse induced by exogenous shocks**, followed by **non-derivable bifurcations**. In these cases, the surviving lineage does not emerge as a deterministic continuation of the previous system, but as a contingent replacement introduced from outside the collapsed domain.

The Last Glacial Maximum (LGM) provides a canonical case. Both Neanderthals and anatomically modern humans occupying Europe prior to the LGM experienced massive population collapse. The subsequent repopulation of Europe by *Homo sapiens* originating from southern refugia, coupled with the persistence of Neanderthal genetic fragments within modern genomes, reveals a structure incompatible with deterministic evolutionary narratives.

2. Conceptual Framework

2.1 Trajectories and Structural Stability

We define an evolutionary *trajectory* as a population-level configuration that remains coherent under a given range of environmental constraints. Stability does not imply optimality, only functional persistence.

2.2 Exogenous Shock

An exogenous shock is an environmental perturbation that exceeds the adaptive envelope of the system. Unlike gradual selective pressures, such shocks do not reward marginal optimization; they invalidate entire modes of existence.

2.3 Structural Collapse

Structural collapse occurs when a system loses the capacity to maintain coherence under altered constraints. Collapse is not equivalent to competitive failure; it is a breakdown of viability.

2.4 Bifurcation and Non-Derivability

A bifurcation is *non-derivable* when the post-event trajectory cannot be inferred from the internal dynamics of the pre-collapse system. In such cases, continuity is imposed externally, not evolved internally.

3. Pre-LGM Stability in Europe

For hundreds of thousands of years, Neanderthals occupied Europe with remarkable ecological stability. Their toolkits, social structures, and physiological adaptations were well matched to Pleistocene environments. Early *Homo sapiens* populations entering Europe during the Upper Paleolithic coexisted with Neanderthals for extended periods.

Crucially, no evidence suggests an intrinsic trajectory toward Neanderthal extinction prior to the LGM. Population structures were stable, and neither species exhibited decisive competitive dominance across the continent.

4. The Last Glacial Maximum as an Exogenous Shock

The LGM (approximately 26,500–19,000 years BP) represents one of the most severe climatic events of the late Pleistocene. Ice sheet expansion, extreme aridity, and ecosystem collapse transformed Europe into a largely uninhabitable domain and archaeological discontinuities lasting thousands of years.

This event functioned not as a selective gradient but as a **structural annihilator**. The adaptive strategies of both Neanderthals and European sapiens populations were rendered ineffective. Survival was not a function of local optimization, but of geographical contingency.

5. Collapse of Local Evolutionary Trajectories

Archaeological and genetic evidence indicates that both Neanderthals and European *Homo sapiens* suffered catastrophic population reductions during the LGM. In many regions, human presence disappears entirely from the record.

This symmetry is critical: the collapse does not reflect competitive displacement. It reflects **system-wide incoherence**. The evolutionary trajectories operating in Europe prior to the LGM ceased to function.

6. Repopulation as Structural Replacement

Post-LGM Europe was repopulated primarily by *Homo sapiens* migrating from southern refugia (e.g., the Balkans, Iberia, and possibly Anatolia). These populations did not descend from the collapsed European groups; they introduced a **new trajectory**.

This repopulation constitutes a **structural replacement**, not an evolutionary continuation. The surviving sapiens lineage was not the endpoint of the prior system—it was an external solution imposed after collapse.

7. Hybridization as Genetic Residue

Modern Eurasian populations carry approximately 1–2% Neanderthal DNA. This genetic material is not evidence of Neanderthal survival as a system. Rather, it represents **residual absorption**.

Key characteristics of this residue:

- It is fragmentary, not systemic
- It persists without preserving Neanderthal social or ecological structures
- It contributes locally advantageous traits (e.g., immune response, keratin production)

When aggregated across the global human population, up to 20–40% of the Neanderthal genome survives in distributed form. This constitutes a **genetic archive** of a collapsed evolutionary structure.

8. The Illusion of Evolutionary Determinism

Deterministic narratives claim that Homo sapiens replaced Neanderthals due to superior cognition or adaptability. This interpretation relies on outcome-based reasoning.

However:

- The LGM killed both populations at similar rates
- Survival depended on geographic refuge, not adaptive superiority
- The repopulating sapiens were not the evolutionary descendants of European sapiens

The outcome is intelligible in retrospect, but it was not derivable from the prior state of the system.

“This does not deny differential cognitive or cultural capacities, but shows that such differences were not sufficient to determine survival under exogenous collapse.”

9. Discussion

This case illustrates a broader principle: **evolutionary history is punctuated by non-derivable bifurcations imposed by structural collapse**. Genes, like institutions or cultures, can survive the destruction of the systems that generated them.

Evolutionary theory must therefore distinguish between:

- Optimization within stable constraints
- Replacement following constraint failure

Failure to do so results in narrative determinism masquerading as explanation.

“This analysis does not reject evolutionary theory, but distinguishes adaptive optimization from structural replacement following systemic collapse.”

“Structural replacement is orthogonal to fitness-based explanation.”

10. Conclusion

The Last Glacial Maximum did not select *Homo sapiens* over Neanderthals. It destroyed both local systems. What followed was not evolutionary necessity, but contingent replacement.

The Neanderthal genetic legacy embedded within modern humans is the physical trace of this bifurcation: a fossil not of bones, but of structure.

History appears deterministic only when the events that broke it are erased.